

Signal Sense – IoT Sensor Anomaly Detection Dashboard (Synthetic IoT Dataset)

Technical Case Study & Domain Research

1. Introduction

The Internet of Things (IoT) connects sensors and smart devices to the internet for collecting and monitoring data. Industries use IoT sensors to measure temperature, humidity, pressure, vibration, voltage, and other parameters.

The **Signal Sense** dashboard uses Machine Learning to analyze sensor data and detect abnormal behavior. This helps industries identify equipment faults early, reduce maintenance costs, and improve operational efficiency through real-time monitoring.

2. Project Domain & Industry Research

Project Domain

- Internet of Things (IoT)
- Industrial IoT (IIoT)
- Smart Sensors
- Embedded Systems
- Machine Learning
- Cloud & Edge Computing

Applications

- Smart Factories

- Manufacturing
- Power Plants
- Healthcare
- Agriculture
- Smart Buildings

Importance

- Continuous monitoring
- Predictive maintenance
- Energy optimization
- Fault detection
- Remote monitoring

3. Existing Systems & Technical Limitations

Existing System	Limitation
Manual Inspection	Slow and error-prone
Rule-Based Monitoring	Cannot detect complex anomalies
Threshold Alerts	High false alarms
Basic Dashboards	No intelligent prediction
Periodic Maintenance	Problems found too late

4. Proposed System

Signal Sense monitors IoT sensor data using Machine Learning to identify abnormal patterns in real time.

Main Modules

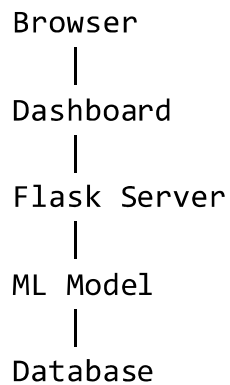
- Sensor Data Collection
- Data Preprocessing
- Machine Learning Prediction
- Anomaly Detection
- Database Storage
- Dashboard Visualization
- Alert Generation

5. Flask Architecture

Components

- Flask Framework
- Routing
- HTML Templates
- Static Files
- Jinja2 Template Engine
- Database
- Machine Learning Model

Architecture



6. HTML, CSS & JavaScript Concepts

HTML

- Forms
- Tables
- Dashboard Layout
- Navigation Bar
- Cards

CSS

- Flexbox
- Grid Layout
- Responsive Design
- Card Styling
- Hover Effects

JavaScript

- Form Validation
- DOM Manipulation
- Event Handling
- Fetch API
- Dynamic Dashboard Updates

7. Database Design

Sensors Table

- Sensor ID
- Timestamp
- Temperature
- Humidity

- Pressure
- Vibration
- Battery Level

Prediction Table

- Prediction ID
- Anomaly Status
- Confidence Score
- Timestamp

Users Table

- User ID
- Username
- Email
- Password

8. Machine Learning Concepts

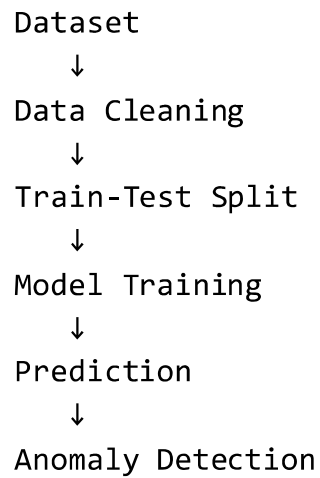
Data Preprocessing

- Data Cleaning
- Missing Value Handling
- Feature Scaling
- Feature Selection

Algorithms

- Random Forest
- Isolation Forest
- Decision Tree
- Logistic Regression
- Support Vector Machine (SVM)
- K-Nearest Neighbors (KNN)

Workflow



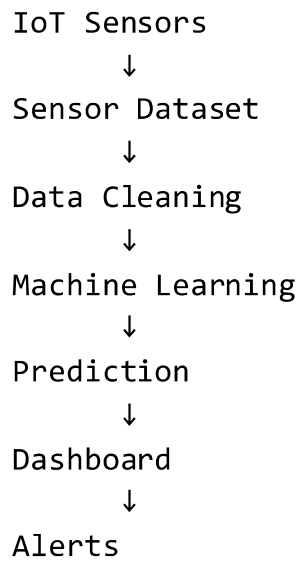
9. Dataset Information

Dataset Name: Synthetic IoT Dataset

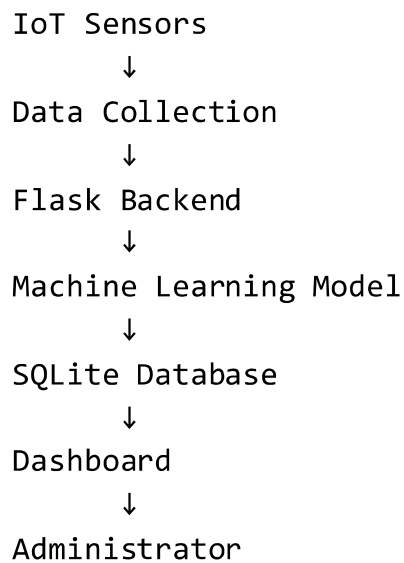
Features

- Sensor ID
- Temperature
- Humidity
- Pressure
- Vibration
- Voltage
- Current
- Power Consumption
- Battery Level
- Timestamp
- Device Status
- Anomaly Label

10. System Workflow



11. Overall System Architecture



12. Real-World Applications

- Smart Factories
- Power Plants
- Oil & Gas Industries
- Hospital Equipment Monitoring
- Data Centers
- Smart Agriculture
- Building Automation

13. Implementation Methodology

1. Requirement Analysis
2. Data Collection
3. Data Cleaning
4. Model Training
5. Flask Development
6. Database Integration
7. Dashboard Development
8. Testing
9. Deployment

14. Advantages

- Early fault detection
- Reduced equipment downtime
- Lower maintenance cost
- Real-time monitoring
- Better resource utilization
- Predictive maintenance
- Improved efficiency

15. Future Enhancements

- Live IoT Sensor Integration
- MQTT Support
- Cloud Deployment
- Mobile Application
- Email & SMS Alerts
- Deep Learning Models
- Edge AI Deployment

16. References

- Synthetic IoT Dataset
- Flask Documentation
- Scikit-learn Documentation
- SQLite/MySQL Documentation
- HTML Documentation
- CSS Documentation
- JavaScript Documentation
- Research papers on IoT, IoT, and Anomaly Detection

17. Reference Images

1. IoT Sensor Network Architecture

